

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12403072
Kind Code	B1
Date of Patent	September 02, 2025
Inventor(s)	Chen; Junyuan

Medication container

Abstract

A medication container includes: a container body, with an accommodation chamber defined therein for receiving medication, wherein a slot is formed at a rear end of the accommodation chamber; and a lid, including a cover plate and an elastic soft rubber insert, wherein the elastic soft rubber insert is inserted into the slot, and a top end of the elastic soft rubber insert protrudes from the slot and is connected to a rear end of the cover plate, such that the cover plate is pivotally connected to the container body only via the elastic soft rubber insert; wherein the cover plate is engaged with an opening of the accommodation chamber by virtue of bending of the elastic soft rubber insert, and the elastic insert is deformed in response to being pulled by the rear end of the cover plate.

Inventors:	Chen; Junyuan (Hong Kong, CN)
Applicant:	AN GROUP LIMITED (Hong Kong, CN)
Family ID:	96435164
Assignee:	AN GROUP LIMITED (Hong Kong, CN)
Appl. No.:	19/212742
Filed:	May 20, 2025

Foreign Application Priority Data

CN	202520845195.3	Apr. 29, 2025
----	----------------	---------------

Publication Classification

Int. Cl.: A61J7/04 (20060101); B65D43/16 (20060101)

U.S. Cl.:

Field of Classification Search

CPC: A61J (7/0481); A61J (7/04); A61J (1/03); A61J (1/1425); B65D (43/163); B65D (43/164)

USPC: 220/829; 220/264; 220/847; 220/827; 220/254.5

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2483304	12/1948	Vogel	220/264	A24F 15/12
6257431	12/2000	Baudin	220/849	B65D 47/0814
6371321	12/2001	Lee	220/23.6	B65D 43/22
7624890	12/2008	Noble	206/538	B65D 83/0445
12139305	12/2023	Leung	N/A	E05D 1/06
2009/0166243	12/2008	Cetera	206/459.5	A61J 1/03
2022/0097949	12/2021	Miura	N/A	A47K 10/421
2023/0202724	12/2022	Chen	220/263	B65D 43/164

Primary Examiner: Impink; Mollie

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to Chinese Patent Application No. 202520845195.3, filed with the Chinese Patent Office on Apr. 29, 2025, titled “MEDICATION CONTAINER”, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to the technical field of storage apparatuses, and in particular, relates to a medication container.

BACKGROUND

(3) To facilitate users in accessing medication, currently, most medication container lids are equipped with an automatic pop-up function. This automatic pop-up function is generally achieved by connecting a torsion spring at a pivot shaft where a lid and a container body are connected. For such configuration, during production and assembly of the medication container, the cost is relatively high and assembly steps are rather cumbersome due to a large number of components or parts.

SUMMARY

(4) In view of the above problem, embodiments of the present disclosure provide a medication container. The medication container, while achieving automatic lid pop-up (or automatic lid ejection), simplifies the structure of the medication container structure, reduces the number of components or parts, and thereby lowers production costs and improves assembly efficiency.

(5) According to one aspect of the embodiments of the present disclosure, a medication container is provided. The medication container includes: a container body, with an accommodation chamber defined therein for receiving medication, wherein a slot is formed at a rear end of the

accommodation chamber; and a lid, including a cover plate and an elastic soft rubber insert, wherein the cover plate and the elastic soft rubber insert are integrally formed, the elastic soft rubber insert is inserted into the slot, and a top end of the elastic soft rubber insert protrudes from the slot and is connected to a rear end of the cover plate, such that the cover plate is pivotally connected to the container body only via the elastic soft rubber insert; wherein the cover plate is engaged with an opening of the accommodation chamber by virtue of bending of the elastic soft rubber insert and is then locked onto the container body, and the elastic insert is deformed in response to being pulled by the rear end of the cover plate; and in a case where the cover plate is unlocked from the container body, the elastic insert recovers from deformation and drives the cover plate to pivot to an open state.

(6) In some embodiments, a first retaining portion is arranged on the elastic soft rubber insert, and a second retaining portion is arranged on an inner sidewall of the slot, wherein the second retaining portion is configured to be engaged with the first retaining portion after the elastic soft rubber insert is inserted into the slot, to restrict the elastic soft rubber insert from sliding outward.

(7) In some embodiments, the first retaining portion includes a plurality of protrusions spaced apart on a rear side surface of the elastic soft rubber insert, and the second retaining portion includes a plurality of recesses spaced apart on a rear inner wall of the slot, wherein the plurality of protrusions and the plurality of recesses are engaged in one-to-one correspondence.

(8) In some embodiments, the plurality of protrusions and the plurality of recesses are all spaced apart along a width direction of the elastic soft rubber insert.

(9) In some embodiments, a width and a thickness of the elastic soft rubber insert are both equal to a width and a thickness of the cover plate.

(10) In some embodiments, the elastic soft rubber insert is configured such that, in an initial state, the top end of the elastic soft rubber insert is bent backward, such that the elastic soft rubber insert is recoverable from deformation to drive the cover plate to pivot by a predetermined angle to be in the open state, wherein the predetermined angle is greater than 90° .

(11) In some embodiments, the lid further includes a seal gasket, wherein the seal gasket is arranged on an inner surface of the cover plate, the seal gasket, the cover plate, and the elastic soft rubber insert form an integral structure, and the seal gasket is configured to abut against an opening edge of the accommodation chamber in a case where the cover plate is engaged with the opening of the accommodation chamber.

(12) In some embodiments, the seal gasket is annular and adapted to the opening edge of the accommodation chamber in terms of shape.

(13) In some embodiments, a front end of the cover plate extends curvedly to form an engagement plate, wherein an engagement portion is arranged on the engagement plate, and an elastic engagement member is arranged on a front end of the container body, wherein the elastic engagement member is configured to be engaged with the engagement portion in a case where the cover plate is engaged with the opening of the accommodation chamber, to lock the cover plate in the closed state.

(14) In some embodiments, the medication container further includes a plurality of storage compartments arranged therein, wherein each of the storage compartments includes the container body and the lid.

(15) In the medication container according to the embodiments of the present disclosure, the lid includes a cover plate and an elastic soft rubber insert that are integrally formed. The cover plate is configured to close or open the accommodation chamber in the container body. The elastic soft rubber insert is inserted into the slot of the container body, enabling the cover plate to pivot relative to the container body through the bending of the elastic soft rubber insert, such that the need for a pivot shaft for rotation between the cover plate and the container body is eliminated, and structural simplification is achieved. At the same time, by virtue of deformation recovery capabilities of the elastic soft rubber insert, the cover plate may automatically pop up after being unlocked, realizing

an automatic pop-up lid function, such that users are facilitated in accessing medication within the accommodation chamber and user experience is ensured.

(16) Furthermore, the medication container according to the embodiments of the present disclosure may, as a whole, consist of only two components or parts: the container body and the lid. A reduced number of components or parts is conducive to lowering production costs. Moreover, the assembly relationship between the container body and the lid is the insertion of the elastic soft rubber insert into the slot. Such an assembly method is easy to operate, such that the assembly difficulty of the medication container is reduced and the assembly efficiency is enhanced.

(17) The above description only summarizes the technical solutions of the present disclosure. Specific embodiments of the present disclosure are described hereinafter to better and clearer understand the technical solutions of the present disclosure, to practice the technical solutions based on the disclosure of the specification, and to make the above and other objectives, features and advantages of the present disclosure more apparent and understandable.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) By reading the detailed description of preferred embodiments hereinafter, various other advantages and beneficial effects become clear and apparent for persons of ordinary skill in the art. The accompanying drawings are merely for illustrating the preferred embodiments, but shall not be construed as limiting the present disclosure. In all the accompanying drawings, like reference numerals denote like parts. In the drawings:

(2) FIG. 1 is a schematic perspective view of a medication container in an open state according to some embodiments of the present disclosure;

(3) FIG. 2 is a schematic exploded view of a medication container in an open state according to some embodiments of the present disclosure;

(4) FIG. 3 is a schematic perspective view of a medication container in a closed state according to some embodiments of the present disclosure;

(5) FIG. 4 is a schematic cross-sectional view of a medication container in a closed state according to some embodiments of the present disclosure;

(6) FIG. 5 is a schematic cross-sectional view of a medication container with a cover plate thereof being in an unlocked state according to some embodiments of the present disclosure;

(7) FIG. 6 is a schematic cross-sectional view of a medication container in an open state according to some embodiments of the present disclosure;

(8) FIG. 7 is a schematic rear view of a lid of a medication container according to some embodiments of the present disclosure;

(9) FIG. 8 is a schematic side view of a lid of a medication container according to some embodiments of the present disclosure;

(10) FIG. 9 is a schematic perspective view of a bottom of a medication container according to some embodiments of the present disclosure; and

(11) FIG. 10 is a schematic perspective view of a medication container according to some embodiments of the present disclosure.

(12) Reference numerals in the embodiments and denotations thereof: **100**-medication container; **101**-storage compartment; **110**-container body; **111**-accommodation chamber; **112**-slot; **1121**-second retaining portion; **11211**-recess; **113**-elastic engagement member; **1131**-support arm; **1132**-press member; **11321**-engagement protrusion; **120**-lid; **121**-cover plate; **122**-elastic soft rubber insert; **1221**-first retaining portion; **12211**-protrusion; **1212**-engagement plate; **12121**-engagement portion; **12122**-slot; **123**-seal gasket.

DETAILED DESCRIPTION

- (13) The embodiments containing the technical solutions of the present disclosure are described in detail with reference to the accompanying drawings. The embodiments hereinafter are only used to clearly describe the technical solutions of the present disclosure. Therefore, these embodiments are only used as examples, but are not intended to limit the protection scope of the present disclosure.
- (14) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the present disclosure pertains. The terms used herein in the specification of present disclosure are only intended to illustrate the specific embodiments of the present disclosure, instead of limiting the present disclosure. The terms “comprise,” “include,” and any variations thereof in the specification, claims, and the description of the drawings of the present disclosure are intended to cover a non-exclusive inclusion.
- (15) In the description of the present disclosure, the terms “first,” “second,” and the like are only used for distinguishing different objects, but shall not be understood as indication or implication of relative importance or implicit indication of the number of the specific technical features, the specific sequence or priorities. In the description of the embodiments of the present disclosure, the term “multiple” or “a plurality of” signifies at least two, unless otherwise specified.
- (16) The terms “example” and “embodiment” in this specification signify that the specific characteristic, structures or features described with reference to the embodiments may be covered in at least one embodiment of the present disclosure. This term, when appearing in various parts of the specification, neither indicates the same embodiment, nor indicates an independent or optional embodiment that is exclusive of the other embodiments. A person skilled in the art would implicitly or explicitly understand that the embodiments described in this specification may be incorporated with other embodiments.
- (17) In the description of the embodiments of the present disclosure, the term “and/or” is merely an association relationship for describing associated objects, which represents that there may exist three types of relationships. For example, the phrase “A and/or B” may indicate (A), (B), or (A and B). In addition, the forward-slash symbol “/” generally represents an “or” relationship between associated objects before and after the symbol.
- (18) In the description of the embodiments of the present disclosure, the term “multiple” or “a plurality of” signifies more than two (including two), unless otherwise specified. Likewise, the term “a plurality of groups” or “multiple groups” signifies more than two groups (including two groups), and the term “a plurality of pieces” or “multiple pieces” signifies more than two pieces (including two pieces).
- (19) In the description of the embodiments of the present disclosure, it should be understood that the terms “central,” “transversal,” “longitudinal,” “length,” “width,” “thickness,” “upper,” “lower,” “front,” “rear,” “left,” “right,” “vertical,” “horizontal,” “top,” “bottom,” “inner,” “outer,” “clockwise,” “counterclockwise,” “axial,” “radial,” “circumferential,” and the like indicate orientations and position relationships which are based on the illustrations in the accompanying drawings, and these terms are merely for ease and brevity of the description, instead of indicating or implying that the devices or elements shall have a particular orientation and shall be structured and operated based on the particular orientation. Accordingly, these terms shall not be construed as limiting the present disclosure.
- (20) In the description of the embodiments of the present disclosure, it should be noted that unless otherwise specified and defined, the terms “mounted,” “coupled,” “connected,” “secured,” and derivative forms thereof shall be understood in a broad sense, which, for example, may be understood as secured connection, detachable connection or integral connection; may be understood as mechanical connection or electrical connection, or understood as direct connection, indirect connection via an intermediate medium, or communication between the interiors of two elements or interactions between two elements. Persons of ordinary skill in the art may understand the specific meanings of the above terms in the embodiments of the present disclosure according to

the actual circumstances and contexts.

(21) To achieve an automatic pop-up lid, in conventional medication containers, in addition to configuring a torsion spring on the pivot shaft, include some medication containers where, based on a pivotal connection between the lid and the container body via a pivot shaft, an elastic soft rubber strip is further arranged on an inner side of the pivotal connection between the lid and the container body, to achieve automatic lid opening by virtue of the elastic deformation of the elastic soft rubber strip.

(22) Regardless of a torsion spring or an elastic soft rubber strip, during the production of the medication container, the torsion spring and the elastic soft rubber strip both need to be produced and processed separately. That is, the torsion spring or the elastic soft rubber strip is first prepared as an independent component or part and then assembled with the container body and the lid to form a complete medication container. This undoubtedly leads to a more complex structure for the medication container and a larger number of components or parts, consequently causing an increase in cost and assembly difficulty. Moreover, during assembly, the container body and the lid still need to be connected via a pivot shaft, which further leads to a decrease in assembly efficiency.

(23) In view of the aforementioned problems, the present disclosure considers improving the medication container by reducing the number of components or parts and simplifying assembly steps, to achieve the objectives of cost reduction and ease of assembly. Through practical research, the present disclosure utilizes conventional processing technologies such as two-shot injection molding or dual-color injection molding to integrally form the cover plate and the elastic soft rubber insert from different materials to form the lid. Simultaneously, a slot is formed at the rear end of the accommodation chamber on the container body, and the lid is assembled with the container body by inserting the elastic soft rubber insert into the slot.

(24) Based on this, the cover plate may pivot relative to the container body through the elastic deformation of the elastic soft rubber insert, thereby achieving the opening and closing of the cover plate. For such configuration, there is no need to configure a pivot shaft to rotatably connect the lid to the container body, such that the structure of the medication container is simplified and the assembly difficulty is reduced. At the same time, by virtue of the elastic recovery capabilities of the elastic soft rubber insert, the cover plate may automatically pivot and pop up in response to being opened, such that users are facilitated in accessing medication and user experience is ensured.

(25) First, referring to FIG. 1 and FIG. 2, which respectively illustrate a perspective view and an exploded view of a medication container **100** in an open state according to some embodiments of the present disclosure, the medication container **100** includes a container body **110** and a lid **120**. The container body **110** defines an accommodation chamber **111** therein for receiving medication. At a rear end of the accommodation chamber **111**, a slot **112** is also formed on the container body **110**. The lid **120** includes a cover plate **121** and an elastic soft rubber insert **122**. The cover plate **121** and the elastic soft rubber insert **122** are integrally formed. As previously described, the cover plate **121** and the elastic soft rubber insert **122** may be processed using existing injection molding techniques such as two-shot injection molding or dual-color injection molding to form an integral structure. The elastic soft rubber insert **122** is inserted into the slot **112**. A top end of the elastic soft rubber insert **122** protrudes from the slot **112** and is connected to a rear end of the cover plate **121**. This configuration allows the cover plate **121** to form a pivotal connection to the container body **110** via the elastic soft rubber insert **122**. That is, the cover plate **121** may pivot relative to the container body **110** through the bending deformation of the elastic soft rubber insert **122**, such that the opening and closing of the cover plate **121** is achieved.

(26) The elastic soft rubber insert **122** may be made of silicone, rubber, silicone rubber, thermoplastic elastomer (TPE), thermoplastic polyurethane (TPU), thermoplastic copolyester (TPC) and polyvinyl chloride (PVC), or the like.

(27) As illustrated in FIG. 3, which illustrates a perspective view of the medication container in a closed state, the cover plate **121** may, through the bending of the elastic soft rubber insert **122**, be

engaged with an opening of the accommodation chamber **111**. Upon engagement, the cover plate **121** may be locked onto the container body **110** by means such as latching or locking, to maintain a sealed protection for the accommodation chamber **111** and achieve reliable storage of medication within the accommodation chamber **111**. In a case where the cover plate **121** is unlocked, the elastic soft rubber insert **122** automatically recovers from deformation. During the process of recovering from deformation, the elastic soft rubber insert **122** drives the cover plate **121**, via its top end, to pivot to an open state.

(28) Specifically, referring to FIG. 4 to FIG. 6, which respectively show cross-sectional views of the cover plate **121** in three states during the process of pivoting from a closed state to an open state, driven by the elastic soft rubber insert **122**. In the state illustrated in FIG. 4, the cover plate **121** is locked onto the container body **110**, and at this time, the cover plate **121** remains in the closed state. On this basis, in a case where the cover plate **121** is unlocked as illustrated in FIG. 5, the cover plate **121**, under an elastic force of the elastic soft rubber insert **122**, may pivot in the direction indicated by the arrow in FIG. 5, achieving an automatic pop-up lid function. After the cover plate **121** pivots to the open state, a state illustrated in FIG. 6 is presented. In this state, the cover plate **121** is maintained under a support effect of the elastic soft rubber insert **122**.

(29) In the medication container **100** according to the embodiments of the present disclosure, the lid **120** includes a cover plate **121** and an elastic soft rubber insert **122** that are integrally formed. The cover plate **121** is responsible for closing and opening the accommodation chamber **111** on the container body **110**. The elastic soft rubber insert **122** is inserted into the slot **112** in the container body **110**, such that the cover plate **121** may pivot relative to the container body **110** through the deformation and recovery from deformation of the elastic soft rubber insert **122**. In a case where the cover plate **121** is in the closed state, the cover plate **121** is engaged with the opening of the accommodation chamber **111** and is locked onto the container body **110**, and the elastic soft rubber insert **122** is deformed in response to being pulled by the rear end of the cover plate **121**. In a case where the cover plate **121** is unlocked from the container body **110**, the elastic soft rubber insert **122** recovers from deformation and drives the cover plate **121** to pivot to the open state.

(30) Through the above-described manner, the pivot shaft required for rotation between the cover plate **121** and the container body **110** may be eliminated, achieving the purposes of structural simplification and improved space utilization. At the same time, by virtue of deformation recovery capabilities of the elastic soft rubber insert **122**, the cover plate **121** may automatically pop up after being unlocked, realizing an automatic pop-up lid function, such that users are facilitated in accessing medication within the accommodation chamber **111** and user experience is ensured.

(31) Furthermore, the medication container **100** according to the embodiments of the present disclosure may, as a whole, consist of only two components or parts: the container body **110** and the lid **120**. A reduced number of components or parts is conducive to lowering production costs. Moreover, the assembly relationship between the container body **110** and the lid **120** is the insertion of the elastic soft rubber insert **122** into the slot **112**. Such an assembly method is easy to operate, such that the assembly difficulty of the medication container **100** is reduced and the assembly efficiency is enhanced.

(32) To ensure the stability of the assembly connection between the lid **120** and the container body **110**, the present disclosure further provides a reinforced design for the connection of the elastic soft rubber insert **122** in the slot **112**. Specifically, still referring to FIG. 2, which also illustrates a perspective view of an inner wall of the slot **112** in the medication container **100**. As illustrated in the FIG. 2, a first retaining portion **1221** is arranged on the elastic soft rubber insert **122**, and a second retaining portion **1121** is arranged on an inner side wall of the slot **112**. After the elastic soft rubber insert **122** is inserted into the slot **112**, the second retaining portion **1121** and the first retaining portion **1221** are engaged with each other, such that the outward sliding of the elastic soft rubber insert **122** is restricted and the elastic soft rubber insert **122** is ensured to be securely connected within the slot **112**.

(33) Specifically, still referring to FIG. 2 and in further combination with FIG. 3, the first retaining portion **1221** may include a plurality of protrusions **12211** spaced apart on a rear side surface of the elastic soft rubber insert **122**, and the second retaining portion **1121** includes a plurality of recesses **11211** spaced apart on a rear inner wall of the slot **112**, wherein the plurality of protrusions **12211** and the plurality of recesses **11211** are engaged in one-to-one correspondence. In the embodiments, by engagement between the plurality of spaced protrusions **12211** with the plurality of spaced recesses **11211**, the container body **110** forms a limit on the elastic soft rubber insert **122** at multiple points. This ensures that when the elastic soft rubber insert **122** recovers from elastic deformation and drives the cover plate **121** to pivot to the open state, the elastic soft rubber insert **122** may not be disengaged from the slot **112** due to an inertial force of the cover plate **121**.

(34) As illustrated in FIG. 2, the plurality of protrusions **12211** may be spaced apart along a width direction of the elastic soft rubber insert **122** (the direction indicated by the double-head arrow W in FIG. 2). This allows the elastic soft rubber insert **122** to be retained at a plurality of positions in the width direction, such that the elastic soft rubber insert **122**, in response to being inserted and fixed in the slot **112**, is less prone to tilting, and thus the closing of the cover plate **121** is not affected.

(35) It is understandable that, for the assembly and fixing of the elastic soft rubber insert **122** in the slot **112**, in addition to the engagement between the first retaining portion **1221** and the second retaining portion **1121**, methods such as interference fit or adhesion may also be employed.

(36) Referring to FIG. 7 which illustrates a rear view of the lid **120** and FIG. 8 which illustrates a side view of the lid, a width W1 of the elastic soft rubber insert **122** may be set equal to a width W2 of the cover plate **121**, and a thickness H1 of the elastic soft rubber insert **122** may also be set equal to a thickness H2 of the cover plate **121**. This configuration, while ensuring the overall flatness and aesthetic appeal of the lid **120**, also ensures that in a case where the elastic soft rubber insert **122** drives the cover plate **121** to pivot from its rear end, the rear end of the cover plate **121** experiences a uniform force along the width direction (i.e., the direction indicated by the double arrow W in FIG. 2). This, in turn, ensures that the cover plate **121** pivots smoothly to the open state without tilting, wobbling, or similar issues.

(37) To ensure that users may more easily access medication in the accommodation chamber **111** after the cover plate **121** is opened, as illustrated in FIG. 8, the dashed line portion in FIG. 8 represents the lid **120** in the closed state, and the solid line portion represents the lid **120** in the open state. In an initial state, as illustrated by the solid line portion in FIG. 8, i.e., in an undeformed default state, the top end of the elastic soft rubber insert **122** is bent rearward (in the leftward direction as illustrated in FIG. 8). This allows the elastic soft rubber insert **122**, after the cover plate **121** is unlocked, to drive the cover plate **121** to pivot by a predetermined angle α greater than 90° from the closed state (illustrated by the dashed line portion) to the open state (illustrated by the solid line portion), and ultimately maintain the cover plate **121** in the open state.

(38) In a deformed state as illustrated in FIG. 8, the top end of the elastic soft rubber insert **122** is upright. Furthermore, in the deformed state, the elastic soft rubber insert **122** may also be pulled by the rear end of the cover plate **121** to be slightly bent forward (in the rightward direction as illustrated in FIG. 8).

(39) Still referring to FIG. 1, in a case where the cover plate **121** pivots by the predetermined angle α , the opening of the accommodation chamber **111** may be fully exposed, such that it is more convenient for the user to reach, with hands, into the accommodation chamber **111** to access medication. In the embodiments illustrated in the accompanying drawings, the predetermined angle α is 120° . In some other embodiments, the predetermined angle α may also be any angle greater than 90° , which is not specifically limited herein.

(40) Moreover, since the top of the elastic soft rubber insert **122** is bent rearward in its initial state, even after long-term use, even though the elastic soft rubber insert **122** fails to fully recover to its initial state due to increased fatigue, the elastic soft rubber insert **122** still ensures that the cover

plate **121** pivots to the open state by a large angle, such that the service life of the medication container **100** is extended.

(41) To better store medication and prevent the medication from becoming damp and ineffective, as illustrated in FIG. 1, the lid **120** may also include a seal gasket **123**. The seal gasket **123** is arranged on an inner surface of the cover plate **121**, and the seal gasket **123**, the cover plate **121**, and the elastic soft rubber insert **122** form an integral structure. This avoids additionally increasing the number of components or parts and affecting assembly efficiency. Similarly to what has been described earlier, the seal gasket **123**, the cover plate **121**, and the elastic soft rubber insert **122** may be integrally formed using conventional processes such as tri-color injection molding. To simplify the production process and reduce costs, the seal gasket **123** and the elastic soft rubber insert **122** may be made of the same material. On this basis, the lid **120** may then be manufactured using simpler dual-color injection molding or two-shot injection molding.

(42) As illustrated in FIG. 4, in a case where the cover plate **121** is engaged with the opening of the accommodation chamber **111**, the seal gasket **123** is abutted against the opening edge of the accommodation chamber **111** (regions A and B outlined in FIG. 4), to achieve sealed protection for the accommodation chamber **111** and ensure that the medication stored inside is less prone to becoming damp and ineffective.

(43) Still referring to FIG. 1, the seal gasket **123** may be annularly arranged and adapted to the opening edge of the accommodation chamber **111** in terms of shape, to provide better sealing for the accommodation chamber **111**.

(44) Regarding the locking of the cover plate **121** onto the container body **110**, the present disclosure proposes an embodiment. Specifically, referring to FIG. 4 and FIG. 9, FIG. 9 illustrates a perspective view of the medication container **100** from a bottom angle. In this embodiment, a front end of the cover plate **121** extends completely to form an engagement plate **1212**, an engagement portion **12121** is arranged on the engagement plate **1212**, and an elastic engagement member **113** is arranged on a front end of the container body **110**. In a case where the cover plate **121** is engaged with the opening of the accommodation chamber **111**, the elastic engagement member **113** is engaged with the engagement portion **12121** to lock the cover plate **121** in the closed state.

(45) Specifically, as illustrated in FIG. 4 to FIG. 6, the engagement portion **12121** may be an engagement recess **12122** formed in the engagement plate **1212**. The elastic engagement member **113** includes a deformably bendable support arm **1131** having a bottom end fixed to a front side of the container body **110**, and a press member **1132** fixed to a top end of the support arm **1131**, wherein an engagement protrusion **11321** is arranged on the press member **1132**.

(46) As illustrated in FIG. 4, after the cover plate **121** is engaged with the opening of the accommodation chamber **111**, the engagement protrusion **11321** is engaged within the engagement recess **12122** to lock the cover plate **121** in the closed state. Based on the state illustrated in FIG. 4, in a case where the press member **1132** is pressed downward in the direction indicated by the arrow in the FIG. 4, the support arm **1131** bends and deforms, causing the engagement protrusion **11321** to move out of the engagement groove **12122**, and presenting the state illustrated in FIG. 5. At this time, the cover plate **121** is unlocked. Subsequently, the cover plate **121**, under the elastic force of the elastic soft rubber insert **122**, may automatically pivot and pop up in the direction indicated by the arrow in FIG. 5 to the open state illustrated in FIG. 6.

(47) Considering that medication intake is generally periodic, to enable users to better distinguish medication required for different time periods, as illustrated in FIG. 10, the medication container **100** may include a plurality of storage compartments **101** arranged therein. Each storage compartment **101** includes the container body **110** and the lid **120**. Specifically, seven storage compartments **101** may be provided as illustrated in FIG. 10, with each storage compartment **101** used for storing medication required for one day. The entire medication container **100** may store medication for one week dose. Additionally, for situations where medication needs to be taken three times a day, three storage compartments **101** may also be provided, each for storing the

medication required for each dose. For other situations, the storage compartments **101** may also be arranged in other quantities, which is not specifically limited herein.

(48) It should be finally noted that the above-described embodiments are merely for illustration of the present disclosure, but are not intended to limit the present disclosure. Although the present disclosure is described in detail with reference to these embodiments, a person skilled in the art may also make various modifications to the technical solutions disclosed in the embodiments, or make equivalent replacements to a part of or all technical features contained therein. Such modifications or replacement, made without departing from the principles of the present disclosure, shall fall within the scope of the present disclosure. Especially, various technical features mentioned in various embodiments may be combined in any fashion as long as there is no structural conflict.

Claims

1. A medication container, comprising: a container body, with an accommodation chamber defined therein for receiving medication, wherein a slot is formed at a rear end of the accommodation chamber; and a lid, comprising a cover plate and an elastic insert, wherein the elastic insert is made of a material selected from the group consisting of silicone, rubber, silicone rubber, thermoplastic elastomer (TPE), thermoplastic polyurethane (TPU), thermoplastic copolyester (TPC) and polyvinyl chloride (PVC), the cover plate and the elastic insert are integrally formed, the elastic insert is inserted into the slot, a top end of the elastic insert protrudes from the slot and is connected to a rear end of the cover plate, and the cover plate is pivotally connected to the container body only via the elastic insert; wherein in a case where the cover plate is in a closed state, the cover plate is engaged with an opening of the accommodation chamber and is locked onto the container body, and the elastic insert is deformed in response to being pulled by the rear end of the cover plate; and in a case where the cover plate is unlocked from the container body, the elastic insert recovers from deformation and drives the cover plate to pivot to an open state.

2. The medication container according to claim 1, wherein a first retaining portion is arranged on the elastic insert, and a second retaining portion is arranged on an inner sidewall of the slot, wherein the second retaining portion is configured to be engaged with the first retaining portion after the elastic insert is inserted into the slot, to restrict the elastic insert from sliding outward.

3. The medication container according to claim 2, wherein the first retaining portion comprises a plurality of protrusions spaced apart on a rear side surface of the elastic insert, and the second retaining portion comprises a plurality of recesses spaced apart on a rear inner wall of the slot, wherein the plurality of protrusions and the plurality of recesses are engaged in one-to-one correspondence.

4. The medication container according to claim 3, wherein the plurality of protrusions and the plurality of recesses are all spaced apart along a width direction of the elastic insert.

5. The medication container according to claim 1, wherein a width and a thickness of the elastic insert are both equal to a width and a thickness of the cover plate.

6. The medication container according to claim 1, wherein the elastic insert is configured such that, in an initial state, the top end of the elastic insert is bent towards a side facing away from the accommodation chamber, such that the elastic insert is recoverable from deformation to drive the cover plate to pivot by a predetermined angle to be in the open state, wherein the predetermined angle is greater than 90°.

7. The medication container according to claim 6, wherein that the elastic insert is configured such that, in the initial state, the top end of the elastic insert is bent by a first angle towards the side facing away from the accommodation chamber, wherein the first angle is 30°, such that the predetermined angle is 120°.

8. The medication container according to claim 1, wherein the lid further comprises a seal gasket,

wherein the seal gasket is arranged on an inner surface of the cover plate, the seal gasket, the cover plate, and the elastic insert form an integral structure, and the seal gasket is configured to abut against an opening edge of the accommodation chamber in a case where the cover plate is engaged with the opening of the accommodation chamber.

9. The medication container according to claim 8, wherein the seal gasket is annular and adapted to the opening edge of the accommodation chamber in terms of shape.

10. The medication container according to claim 1, wherein a front end of the cover plate extends curvedly to form an engagement plate, wherein an engagement portion is arranged on the engagement plate, and an elastic engagement member is arranged on a front end of the container body, wherein the elastic engagement member is configured to be engaged with the engagement portion in a case where the cover plate is engaged with the opening of the accommodation chamber, to lock the cover plate in the closed state.

11. The medication container according to claim 10, wherein the engagement portion is an engagement recess formed in the engagement plate, and the elastic engagement member comprises a support arm having a bottom end fixed to a front side of the container body and being deformably bendable, and a press member fixed to a top end of the support arm, wherein an engagement protrusion is disposed on the press member, and in the case where the cover plate is engaged with the opening of the accommodation chamber, the engagement protrusion is engaged within the engagement recess to lock the cover plate in the closed state.

12. The medication container according to claim 1, further comprising: a plurality of storage compartments arranged therein, wherein each of the storage compartments comprises the container body and the lid.
